Ahmedabad University

# Weekly Report 1

# Disease Prediction System using Machine Learning

Submitted to faculty: Mehul Raval

Student: Ayushi Patel

Enrollment No. : AU2519010

Course: CSE 620 Machine Learning

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# Week 1 – Problem Definition and Dataset Identification

## Work Completed

During the first week, the project's research direction and scope were defined. The objective of the study is to develop a machine learning model capable of predicting diseases based on input symptoms. The motivation is to support preliminary medical diagnosis through data-driven approaches.

A review of publicly available symptom disease datasets was conducted to identify a dataset suitable for supervised learning. After evaluating dataset completeness and structure, an appropriate dataset containing symptom features and corresponding disease labels was selected.

The overall system pipeline :

- Dataset preparation
- Feature representation of symptoms
- Machine learning model training
- Prediction generation

## Tools / Environment

- Python
- Jupyter Notebook
- Git Hub repositories

## Challenges

Identifying a dataset with well-structured symptom attributes suitable for direct machine learning training.

## Plan for Next Week

- Perform detailed dataset exploration.
- Analyse feature distribution and dataset structure.
- Identify potential preprocessing requirements.